Department of Economics

MSc Extended Essay Cover Sheet

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COURSE CODE	**E**	**C**	4	2	3

Did Covid change the use of job search channels ? Evidence from France.

WORD COUNT	**5**	**6**	**5**	**2**

Abstract

This essay looks at whether Covid has changed the way unemployed jobseekers look for work. We use a very rich dataset on the French job market. Two aspects of the job search are considered: the intensity of the job search (number of search channels) and the content of the job search (choice of search channels). In particular, our study focuses on changes in both socio-demographic and previous employment history determinants. We show that there is a general trend towards a reduction in active job search, which has been associated since the Covid period with a reduction in the number of channels used (lower search intensity). Regarding job search content, Covid has the most positive effect on the use of test or job interview and steps to become self-employed, the most negative for the contact with an employer. A more detailed study of the determinants does not reveal any significant cross-effects. However, it raises concerns about the adaptation of the long-term unemployed people to the changes in practices that occurred during the pandemic.

Introduction

Covid-19 seems to have deeply changed the way people work. Studies have already shown the extent of the phenomenon, both for the labour market (demand/supply, unemployment) and working patterns (lockdowns, remote work) [Zarifhonarvar, 2022]. It appears also that Covid effects are heterogeneous across populations. For example, people on the margins of the labour market seem less likely to identify and target new sectors that are doing well [Carrillo-Tudela et al., 2023]. It makes it harder for them to find a job and more generally complicates matching labour supply and demand. Is it just a targeting error? Or has Covid also changed what makes a job search process successful, excluding some population groups to a greater extent? This essay therefore looks at the job search process, via job search behaviour, and in particular job search channels. More specifically, we attempt to answer the following question: has Covid changed the way job-seekers choose and use job search channels? If so, do we observe heterogenous changes depending on job seekers' characteristics ?

Knowing the determinants of job search behaviours is essential for policy-makers to give effective support to job seekers. Policies to get people back to work need to identify levers that increase the efficiency of matches in the labour market, and target groups in the population that require specific support (according to age, level of education, family composition, etc.) to engage into an active job search. In particular, in the long term, it seems essential to measure the changes in job-seeking patterns brought by the pandemic, to adapt existing job-seeking support to new practices in job-search.

Our work follows Bachmann & Baumgarten [2013], who investigate the job search behaviour of unemployed individuals in Europe, and Mussida and Zanin [2020], who study job search channels choices of the unemployed in Italy. We rely on a very rich dataset on the French labour market for the period 2014 – 2022, which does not seem to have been used before to study the impacts of

Covid on the job search. Because of the diversity of job search practices, we restrict this analysis to job seekers who are currently unemployed. These latter are more likely to face long-term unemployment and poverty, as they risk being increasingly excluded from the labour market if they are unable to adapt to changing practices.

First, we construct an indicator of search intensity (number of channels used). We then specify a linear regression model and a probit model to study the effect of Covid on the intensity and content of the job search. Furthermore, we detail the effect of Covid on the socio-demographic characteristics of unemployed job-seekers according to the channels used.

The essay develops as follows. Section 1 provides background information on job search behaviour, efficiency of job search channels, and determinants of job search channel choices. Section 2 describes our data and model specifications. Section 3 presents the results of our analyses. Section 4 concludes and offers some perspectives on further research.

1. Literature review

1.A Different job search behaviours as an outcome of rational choices

Individuals seeking employment influence the search outcome by deciding how much (time, actions, effort) they wish to invest in the process, and how (search channels) [Petrongolo and Pissarides, 2001]. The analysis of these job search behaviours and their consequences on the finding of employment draws heavily from the literature of sociology and psychology.

First, we define what is a job search behaviour. As quoted in Van Hoye's review [2017], Schwab, Rynes, and Aldag [1987] define job search behaviour as the 'sources used to acquire information about job vacancies as well as the intensity with which such information is pursued'. In this work,

our analysis focuses more on the informational stage, via channels to gather information or spread the search, rather than on the use of this information.

Why do we observe different job search behaviour? Unemployed people engage in job search by mobilising their experience (or beliefs) of the labour market, claiming their ability to select methods that work best [Demazière, 2017]. Indeed, each subset of the labour market has specific rules [Rémillon & Marchal, 2012], which is one of the reasons why we observe heterogeneous job search practices. More generally, job search is influenced by all the 'antecedents' of an individual: personality traits, expectations, financial constraint, self-evaluations, motives, social context, and biographical variables [Kanfer, Wanberg, & Kantrowitz, 2001].

Then, based on this knowledge and ‘antecedents’, economics job search theory suggests that unemployed people compare rationally search costs, returns to search effort, and outside options when choosing job search methods [Miano, 2023]. In particular, economists have extensively analysed job search behaviour through effort and intensity, which make it possible to quantify a commitment to job search and partly represent its cost. Various measures have been proposed to quantify this effort. For example, Blau [1994] proposes a measure based on the actions taken by job seekers and their frequency. Other measurement methods are the number of applications during a period [Van den Berg & Van der Klaauw, 2006] or, and this is what we will use here, the number of search methods [Bachmann & Baumgarten, 2013].

1.B Job search channels and their influence on employment outcomes

1.B.1 Typology of job search channels

Job search content is the translation of job search behaviour into actions: this is concrete activities carried out by the job seeker. Individuals can mobilise different search channels, divided into two main categories: formal (job ads, public and private agencies, applications) and informal

(personal network) [Van Hoyle, 2017]. Channels are not perfect substitutes, as they give access to different opportunities, for search as well as public policy. For example, private agencies can act as 'matchmakers' for recruitment programs for disadvantaged groups [Ingold & Valizade, 2017].

Informal search channels involve a wide variety of private contacts, such as friends, family, or former colleagues, and have been the subject of a vast literature. While weak ties greatly extend the number of people someone can know, giving him or her access to more job opportunities [Gravonetter, 1973], it is the so-called strong ties (friends, family) that play an essential role in the job search. These may be the parents of a young graduate [Kramarz & Nordstrom Skans, 2014] or a network of close friends [Cappellari & Tatsiramos, 2010].

Finally, the use of online social networks in job searches is a new field of investigation. Social networks make it easier to maintain a large network of contacts, whether professional or personal, weak or strong [see Garg & Telang, 2018].

1.B.2 Efficiency of job search channels

First, note that the success of a job search is not limited to finding a job, and includes many other dimensions (salary, number of offers, interest, etc.) [Van Hooft, 2021]. The traditional assumption is that the more intensively individuals search for a job, the more information and opportunities they generate [Kanfer, 2001]. However, recent research suggests that it is the quality of the search, rather than its intensity, that is crucial in the relevance of the outcome beyond finding a job [Van Hooft, 2013], hence the interest in studying job search channels in detail, as they are not perfect substitutes.

Indeed, generally speaking, search channels are not equally effective. One of the first studies on this topic was that of Holzer [1988], who showed that the most productive search methods in terms of job offers were the mobilisation of relatives and direct applications. Then, the importance

of informal networks has been highlighted by several empirical studies, such as in Germany [Caliendo et al., 2011] and Italy [Pistaferri, 1999]. About 50% of individuals hear about jobs thanks to their social networks [Moreno Galbis et al., 2020]. However, these informal networks, although widespread and increasing the probability of job offers, do not always lead to the best results, for example regarding the salary obtained, and may even lead to mismatches between workers and jobs' characteristics [Meliciani & Radicchia, 2011].

More generally, beyond the choice of a search method, Weber and Mahringer [2008] show that the outcome of a job search depends very much on individual characteristics. In Austria, for example, a negative selection of workers leads public employment offices to specialise in the support of low-quality workers, for whom they are as effective as other search channels, whereas they produce poor results for jobseekers with higher qualifications. Above all, this leads companies not to offer their 'best offers' to public agencies, thus limiting the possibilities for individuals choosing this search channel.

1.C Determinants of job search channels choices

It is precisely these last two points that interest us in this work. As we have mentioned, not all job search channels are equally successful. Yet, these channels are chosen in heterogeneous ways by job seekers. It is therefore essential to highlight the determinants (gender, education, age, etc.) of this choice, which are also determinants of the return to employment and search outcome. Some empirical studies even suggest that demographic variables are the main determinants of (re)employment and that theoretical frameworks overstate the importance of job-seeking activity [Sverko et al., 2008].

First of all, people who look for a new job while working are more successful in finding a job than the unemployed. An important question is why this difference exists [Blau & Robbins, 1990].

One explanation is that employed people search more effectively because they have better resources (e.g. better contacts) or because they do not suffer from unemployment discrimination. However, another explanation is based on unobserved heterogeneity. In other words, employment status may be a signal of search aptitude or ability. In this case, unobserved differences between individuals are responsible for the lower job search success of the unemployed. In reality, there is little evidence that unemployed and employed jobseekers are similar or apply for the same jobs, As a consequence, it is unlikely that unemployed and employed jobseekers compete directly for jobs (individual characteristics, working time preferences, job search strategies, work history...) [Longhi & Taylor, 2014]. To have a coherent category of job seekers, we then restrict our analysis to unemployed jobseekers.

Having made this distinction, we now summarise the main results of the two papers in line with which we are working. These results will be explored in more detail later, to put our models and the potential effect of Covid into perspective. When it comes to age, formal channels are less likely to be used by younger jobseekers than older ones [Mussida & Zanin, 2020], while the number of job search channels used decreases with age [Bachmann & Baumgarten, 2013]. A higher multichannel attitude is found among men and the most highly educated. Besides, it appears that household characteristics are important predictors of low search intensity, depending on the number of children and elderly dependents [*ibid*]. Finally, for jobseekers with previous job experience, the sector of the previous job affects the choice of the number of search channels as well as the type of channel [Mussida & Zanin, 2020].

To conclude, we mention two articles on job search in France. Ghrairi [2013] studies young people's choice of job search channels, using the same dataset as we do here. Young graduates access their first jobs often through direct applications and with the help of their social network, while the level of education increases the probability of finding a job through professional contact.

Next, Galbis et al [2020] suggest that finding a job through social connections decreases with the job search rate, and they test this prediction with immigrants' job search. In addition, social contacts would be less useful in finding a job during a period of economic expansion.

2. Methodology

1.A Data

The data used in this essay comes from the Continuous Employment Survey (*Enquête emploi en continu*) conducted by the INSEE, the French statistics institute. It is the French version of the Labour Force Survey (LFS) defined at the European level. This survey aims to provide a picture of the activity, employment, and unemployment situation in France. The dataset is very rich and includes socio-demographic variables as well as part of the respondents' employment history. Data are available from 2011 to 2022, but only those for the period 2014-2022, which are harmonised, are used here. However, there is no identification of the individuals taking part in the survey, so it is not possible to cross-check observations by individual or household.

2.A.1 <u>Covid indicator</u>

The Continuous Employment Survey is carried out over the whole year, and there is no indication of the date of data collection. The following approach is proposed to situate the data with the pandemic.

To start with, the years 2014 to 2019 are the 'Pre-Covid' years. Covid restrictions then began in France in mid-March 2020, with the announcement of lockdown on March 16th, 2020. It means that around a quarter of the answers to the 2020 survey were collected before the pandemic had any impact on daily life, and in particular on unemployment and job-seeking. As a consequence, a quarter of the 2020 observations are randomly allocated to the 'Pre-covid' period and the remaining

three quarters are allocated to the 'Covid period'. This randomisation is achieved by drawing a uniform distribution between 0 and 1. Besides, subsidies to companies in the sectors most affected by the pandemic were reduced from July 2021 until the end of January 2022. The whole year 2021 therefore belongs also to the 'Covid period' category. Finally, 2022 is the 'Post-Covid' year.

2.A.2 Unemployed actively searching a job

Our analysis focuses on unemployed people who are actively looking for work, whether or not they have previous work experience. To do this, we use the value associated with the statement: "The individual has made at least one job-seeking activity in the last 4 weeks. ", the answer "No" is the outside option. As mentioned, job searches are very different depending on whether the individual is still employed or not [Wanberg et al, 2019], and this allows us to focus on people who are further away from employment. Over the period 2014-2022, this represents 22,479,302 individuals (for 166,522 observations as this is a weighted dataset). Importantly, employment has remained fairly similar in France between 2014 and 2022, between 10.5% and 7.1%.

In addition, the survey includes data on the socio-demographic situation of the respondents, in particular family organisation, level of education, and age. It also provides information on employment history, such as length of time unemployed or previous employment experience. Table 1 provides descriptive statistics of the samples for the variables we will use, distinguishing between our three Covid periods.

We note slight changes in our sample before and after the pandemic. There are as many men at all periods. The distribution of family set-ups is also almost identical between the three periods, as well as the distribution of education levels. However, there is a greater weighting of the most highly educated from the Covid period onwards (high school, two-year higher education, and particularly bachelor's degree and above), in parallel with a lower representation of the least educated. The biggest change is for age. The percentage of 15-24 year olds falls by a third and the

over 65s almost disappear between pre and post Covid periods. There is a 10% reduction in the proportion of very long-term unemployed (2 years and more) before and after Covid, but a sharp increase in the proportion of those unemployed for less than 3 months. These latter rises from 21% before the Covid, to 30% during the Covid period and up to 39% afterwards.

	Unemployed people that have looked for a job during the last 4 weeks		
	Pre-Covid	**Covid period**	**Post-Covid**
Individual characteristics			
Sexe			
Male	53%	52%	53%
Age range			
15 - 24 years old	37%	30%	*25%*
25 - 49 years old	43%	48%	52%
50 - 64 years old	17%	20%	22%
65 years old or more	3%	2%	1%
Highest education degree			
Primary education	23%	18%	17%
Middle school	8%	7%	8%
Vocational high school	26%	23%	22%
High school	21%	23%	25%
Two-year higher education	9%	10%	10%
Bachelor's degree and above	13%	18%	18%
Family setup			
Single person	18%	20%	19%
Single parent family	18%	18%	19%
Couple without children	14%	15%	14%
Couple with children	43%	41%	40%
Other	6%	6%	7%
Employment history			
Length of time unemployed			
Less than 3 month	21%	30%	39%
3 - 6 months	17%	18%	16%
6 - 12 months	18%	18%	16%
12 - 24 months	21%	18%	15%
2 - 4 years	10%	8%	8%
More than 4 years	14%	8%	6%
Previous job experience			
Yes	84%	87%	88%

Table 1 – Summary statistics of the characteristics of unemployed people looking for a job, before during and after Covid

2.A.3 Job search channels

The search channels present in our dataset are summarised in Table 2. Each is a dummy variable. Note that every unemployed person who said they were looking for work answered 'yes' for at least one search channel (consistency between the different variables of the dataset). Additionally, using Bachmann and Baumgarten's [2013] method, we compute a search intensity variable. It is defined as the number of different channels used by an individual over the last 4 weeks. Finally, Table 2 also includes the "no search" variable, which corresponds to people who are unemployed but have not looked for work in the last four weeks.

Variable label	Definition	Use across sample		
		Pre Covid	Covid	Post Covid
National employment agency	Has contacted the public employment agency	64.8%	63.8%	66.1%
Private employment agency	Has contacted a private employment agency	39.3%	39.2%	40.5%
Personal network	Has asked their personal network (family, friends, colleagues, trade unions, etc.) to help them	65.0%	62.6%	60.2%
Job offers study	Has studied job offers	86.2%	85.7%	84.4%
Competitive exam for the public sector	Has passed a competitive exam to join the civil service	1.3%	1.1%	1.23%
Test or job interview	Had a test or a job interview	19.6%	23.0%	26.7%
Contact with an employer	Contacted an employer either at a job fair, unsolicited application, or in response to a head-hunter	63.7%	46.1%	36.4%
Steps to become self-employed	Has taken steps to become self-employed	2.6%	6.4%	9.2%
Post of a job search/CV (inc, newspaper, online)	Has posted a job search/CV (inc. newspaper, online)	63.8%	67.3%	66.2%
No search		5.2%	7.34%	8.5%
Intensity	Number of search channels used during the last 4 weeks (mean and standard error)	3.7 (1.78)	3.5 (1.83)	3.5 (1.84)

Table 2 – Sample repartition of use of search channels and intensity of search, before during and after Covid

The most commonly used channel is looking at job offers, which is the most immediate when you think of looking for a job. The proportions are stable pre- and post-Covid for most of the search channels, except for contacting an employer and becoming self-employed. We also note slight changes in using one's personal network and doing tests or job interviews, as well as a rise in the proportion of unemployed people that are not job seekers. Finally, the average intensity is fairly high, with jobseekers declaring that they have used an average of between 3.5 and 3.7 different search channels during the last 4 weeks.

There is no strong correlation between our job search channels (Table 3). The negative coefficients in line (8) suggest that becoming self-employed would be a substitute for other job search channels. The other job search channels would, to some extent, be complements to each other.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
National employment agency (1)	1.000								
Private employment agency (2)	0.155*	1.000							
Personal network (3)	0.096*	0.106*	1.000						
Job offers study (4)	0.019*	0.021*	0.023*	1.000					
Competitive exam for the public sector (5)	0.194*	0.198*	0.175*	0.007*	1.000				
Test or job interview (6)	0.090*	0.132*	0.100*	0.080*	0.150*	1.000			
Contact with an employer (7)	0.114*	0.184*	0.198*	0.043*	0.296*	0.214*	1.000		
Steps to become self-employed (8)	0.012*	-0.059*	0.037*	0.004	-0.129*	-0.010*	-0.082*	1.000	
Post of a job search/CV (inc, newspaper, online) (9)	0.158*	0.220*	0.183*	0.041*	0.412*	0.260*	0.364*	-0.024*	1.000

** p<0.01*

Table 3 – Correlation matrix of job search channels use in our sample

2.B Model specification

As a reminder, we are looking for changes in job-seeking behaviour since the Covid pandemic. All our specifications here will only allow us to take an early look at the changes in practices following the Covid pandemic. As we have little data for the post-Covid period, it will be difficult to know whether certain trends observed are not simply transitory, and our analysis obviously needs to be extended when more data becomes available.

We are interested in this change in terms of two components:

- The effect of Covid on the intensity of the job search via the number of search channels
- The effect of Covid on the content of the job search via the choice of search channels.

If there is an effect, we will then try to see how it breaks down on the determinants of the job search. We therefore begin with a descriptive analysis, which we will seek to confirm in our

modelling. This modelling is inspired by the study Determinants of the Choice of Job Search Channels by Mussida and Zanin [2020], which focuses on Italy. These determinants, which will also be our control variables, are :

- Gender
- Age range
- Highest degree
- Previous job experience
- Length of time unemployed

2.B.1 Effect of Covid on job-search intensity

Our variable of interest is *Intensity*. It is constructed as the number of search channels used by an individual over the last 4 weeks. Hence intensity is an integer between 0 and 9 for any individual *i*, any year *t*. It is a count variable, which is often modelled in this context by a Poisson regression [Nyarko et al., 2014; Mussida and Zanin, 2020]. Here, however, our variable is not concentrated downward around 0 and the lowest values, as shown in Figure 1. Furthermore, we see in Table 2 that the mean is twice as large as the variance, contrary to the assumptions of the Poisson model. We therefore choose a linear regression[1]. As Covid is a linear combination of year dummies, we only allow a time-trend and not a year fixed-effect.

Finally, the relationship we are studying is as follows, with β_3 our coefficient of interest and X_{it} the determinants introduced before.

$$Intensity_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 Years_t + \beta_3 Covid_t + \epsilon_{it} \qquad (1)$$

Next, to be able to study the effects of Covid on the socio-demographic determinants of job search in more detail, we run new regressions using interactions between our variables. Thus, we get all

[1]This choice of specification was also guided by a presentation of Ordinary Least Squares and Poisson Regression models by Luc Anselin [in Levine, 2015]

the cross-effects of Covid on the determinants of job search. This is the same specification as (1) but with interaction terms:

$$Intensity_{it} = \alpha_0 + \alpha_1 X_{it} + \alpha_2 Covid_{it} + \alpha_3 X_{it} * Covid_t + \alpha_4 Years_t + \epsilon_{it} \qquad (2)$$

2.B.2 Effect of Covid on job-search content

We use the same socio-demographic variables X_{it}, but this time we break down the potential effect of the pandemic on each of the job-search channels. As our dependent variables are binary, we choose to conduct a probit regression.

$$Channel_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 Years_t + \beta_3 Covid_t + \epsilon_{it} \qquad (4)$$

First, we identify the search channels for which the effect of the Covid is strongest (magnitude of coefficient, significance). For these, we then detail the effect of Covid on the socio-demographic characteristics of the job-seekers using these channels. We therefore adapt equation (4) as follows:

$$Channel_{it} = \alpha_0 + \alpha_1 X_{it} + \alpha_2 Covid_{it} + \alpha_3 X_{it} * Covid_t + \alpha_4 Years_t + \epsilon_{it} \qquad (2)$$

This is again a probit regression.

Of course, our specifications here may suffer from endogeneity bias as many explaining variables of job search behaviours are left apart (access to public transport, employment sector, health, etc.). It will therefore be difficult to conclude on a possible causal effect of Covid beyond an observation of some changes in trend.

3. Results

3.A Descriptive analysis

Figure 1 shows the distributions of the *intensity* variable for each of the three Covid periods.

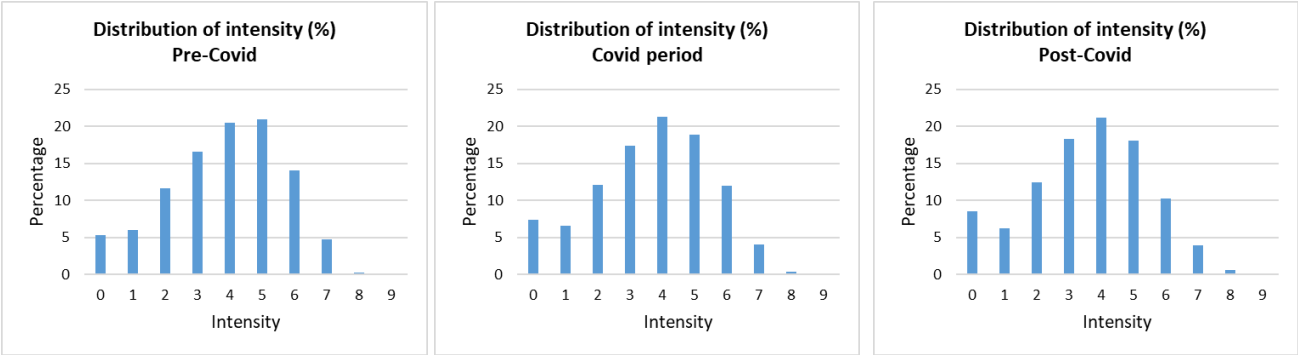

Figure 1 – Distribution of job-search intensity over Covid periods

The frequency of the 0 value increases over time, which is consistent with the evolution over time of the number of unemployed people actively looking for a job (Figure 2). In addition, we see a gradual shift in the intensity distribution towards the left, with an increase in the frequency of values 1, 2, 3, and 4 alongside a decrease in the frequency of intensities 5, 6, and 7. The gap is more pronounced between pre- and post-Covid, with an intermediate gap for the Covid period. It suggests a gradual shift, although we do not know at this stage whether it began at the time of Covid or reflects a longer trend.

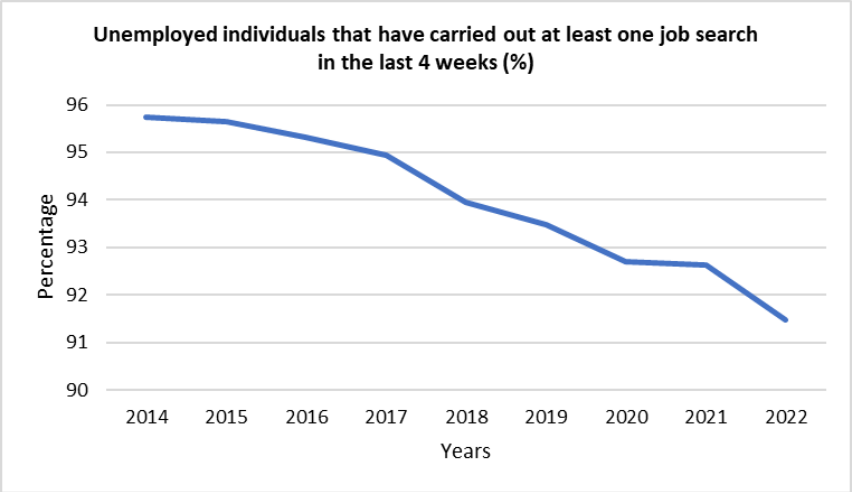

Figure 2 – Evolution of the percentage of unemployed people actively searching for a job between 2014-2022

Figure 2 represents the proportion of unemployed people who are actively looking for work, i.e. unemployed people who say they have made at least one job search in the last four weeks. There has been a constant decline in this proportion over time. Almost 96% of unemployed people were actively looking for work in 2014, whereas this figure falls to 91% in 2022. Despite a stagnation

in 2021, the post-covid years seem to be in line with the general trend. This is consistent with Figure 1.

Last, Figure 3 describes the use of job search channels over time. The year 2020 marks a shift in several trends. The percentage of people who decide to become self-employed increases by more than 5% between 2020 and 2021, tripling. Other small increases can be seen after 2020:

- The one of unemployed people taking a test or a job interview
- The one of unemployed people turning to the national employment agency

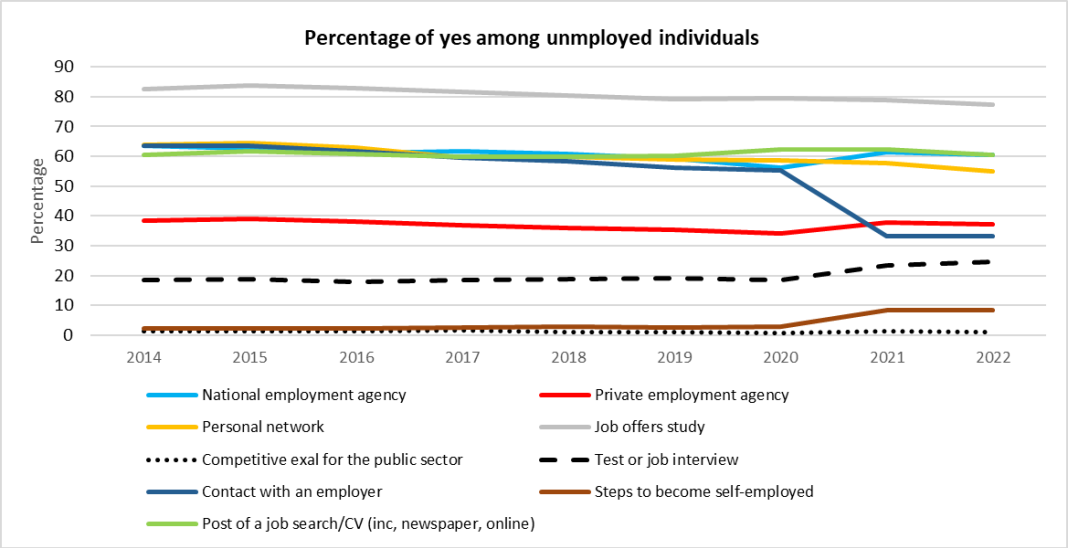

Figure 3 – Descriptive evolution of the use of search channels by unemployed people looking for a job

Conversely, direct contact with an employer logically falls sharply in 2021, a year in which many job fairs or meetings have been cancelled, but this seems to be confirmed in 2022. More slightly, fewer people are using their personal network. There seems to be a drop in the percentage of people studying job offers. However, as this definition is very general and the proportion of "yes" responses is high (around 80%), studying job offers seems to be equivalent to actively looking for a job. Thus, it is not surprising to see the same trend as shown in Figure 2 for actively seeking employment.

In conclusion, there does seem to be an effect of Covid on the content and intensity of the job search, but care must be taken to distinguish this effect from a more general time trend.

3.B Determinants of job search intensity

Results of OLS regression of job-search intensity on socio-demographic characteristics and Covid periods are reported Table 4. Being in the post-Covid period has a negative and significant effect on job-search intensity. The effect is of the same order of magnitude as that of the time trend (coefficient of Year), even if this magnitude is rather low compared to the other variables included in our model. Note that the negative coefficient associated with the time-trend suggests a general trend toward a reduction in the number of channels used, as discussed in Section 3.A.

Variable	Intensity	Variable	Intensity
Covid		Previous job experience	
Pre-Covid	(base)	Yes	(base)
Covid period	-0.0168***	No	-0.3557***
Post-Covid	-0.0145***		
		Family setup	
Sexe		Single person	(base)
Men	(base)	Single parent family	-0.0150***
Women	-0.1512***	Couple without children	-0.0057***
		Couple with children	-0.0175***
Age range		Other	-0.1596***
15-24 y.o	(base)		
25-49 y.o.	-0.1787***	Unemployment length	
50-64 y.o.	-0.3544***	Less than 3 months	(base)
65 y.o. or more	-0.8010***	3-6 months	0.2104***
		6-12 months	0.1923***
Education		12-24 months	0.0767***
Primary education	(base)	2-4 years	-0.0929***
Middle School	0.2097***	More than 4 years	-0.3523***
Vocational high school	0.3042***		
High School	0.4466***	Year	-0.0360***
Two-year higher education	0.6488***		
Bachelor degree or above	0.6237***		
Constant	76.59***		
R^2	0.0595		
Observations	22,474,478		

Legend: * p<0.05; ** p<0.01; *** p<0.001

Table 4 – OLS estimates of the determinants for job-search intensity (number of channels used)

In Table 5, being in the pre-Covid period is the base for cross-effects. Interestingly, gender bias diminishes by 2/3 after Covid. Age reduces the number of channels used (consistent with the literature), increasing in magnitude with age category. Being in the post-Covid period strongly

reduces the negative effect of age, as shown by the positive coefficients of the interaction with Post-Covid for the 25-49 and 50-64 years old groups. One point of attention is the over-65s, who are less active in their post-Covid search than in pre-Covid (significant at the 5% level). A less intense job search, or even a drop of the search, by older jobseekers is a common consequence of their discouragement in the face of discrimination in recruitment [Watermann et al., 2023]. If this effect is amplified as a result of the pandemic, there is a risk that older people will be even further away from a return to employment, with early forced retirements.

The most surprising effect is what happens with the family set-up, with single individuals being the reference category. We have what we would expect for the pre-Covid variables: the presence of children reduces search intensity, and this effect is even stronger if it is a single parent. However, post-Covid, we observe exactly the opposite. Being part of a single-parent family now increases the number of job search channels used (significant at the 5% level). However, other coefficients are not significant. This may be due to the short time perspective we have already mentioned, and more data is needed to investigate further. Nor do we observe any lasting cross-effect of Covid on the duration of unemployment, with a probably transitory increase during the covid period which disappears in the post-Covid period. Finally, the effect of a previous job experience on job-search intensity is reduced by a third in the post-Covid years (significant at the 5% level).

To sum up, it seems that Covid accelerates a general time-trend of a reduction in the number of channels used by job-seekers. The marginal effect of the determinants studied here also decreases post-Covid with cross-effects of opposite signs compared to the pre-Covid period.

Variable	Intensity	Variable#Covid	Intensity
Covid status			
Pre-Covid	(base)		
Covid period	-0.095		
Post-Covid	-0.2569**		
Sexe			
Men	(base)	Women#Covid period	0.0664*
Women	-0.1726***	Women#Post-Covid	0.1094**
Age range		25-49 y.o.#Covid period	0.0887*
15-24 y.o	(base)	25-49 y.o.#Post-Covid	0.2755***
25-49 y.o.	-0.2141***	50-64 y.o.#Covid period	0.0972*
50-64 y.o.	-0.3871***	50-64 y.o.#Post-Covid	0.2304***
65 y.o. or more	-0.7954***	65 y.o. or more #Covid period	-0.068
		65 y.o. or more #Post-Covid	-0.235*
Education		Middle School #Covid period	-0.019
Primary education	(base)	Middle School #Post-Covid	-0.1896*
Middle School	0.2279***	Vocational…#Covid period	-0.024
Vocational high school	0.3122***	Vocational…#Post-Covid	-0.106
High School	0.4605***	High School#Covid period	-0.022
Two-year higher educa..	0.6963***	High School#Post-Covid	-0.1382*
Bachelor degree or ab..	0.6445***	Two-year higher…#Covid period	-0.1632**
		Two-year higher…#Post-Covid	-0.2425***
		Bachelor degree...#Covid period	-0.040
		Bachelor degree...#Post-Covid	-0.2120***
Family setup		Single parent…#Covid period	0.058
Single person	(base)	Single parent…#Post-Covid	0.1673**
Single parent family	-0.0369*	Couple without…#Covid period	-0.041
Couple without children	-0.003	Couple without…#Post-Covid	0.033
Couple with children	-0.011	Couple with…#Covid period	-0.053
Other	-0.1686***	Couple with…#Post-Covid	0.047
		Other#Covid period	0.028
		Other#Post-Covid	0.060
		3-6 months#Covid period	0.070
Unemployment length		3-6 months#Post-Covid	0.1327*
Less than 3 months	(base)	6-12 months #Covid period	0.049
3-6 months	0.1785***	6-12 months#Post-Covid	0.081
6-12 months	0.1686***	12-24 months #Covid period	0.0904*
12-24 months	0.0468**	12-24 months#Post-Covid	0.085
2-4 years	-0.1079***	2-4 years#Covid period	-0.026
More than 4 years	-0.3659***	2-4 years#Post-Covid	0.125
		More than 4 years #Covid period	0.018
		More than 4 years #Post-Covid	0.081
Previous job experience			
Yes	(base)	No#Covid period	0.009
No	-0.3687***	No#Post-Covid	0.1135*
Year	-0.0363***		
Constant	77.3279***		
R^2	0.061		
Observations	22,474,478		

Legend: * p<0.05; ** p<0.01; *** p<0.001

Table 5 – OLS estimates of the determinants for job-search intensity including cross-effects with covid status

3.C Determinants of job search content

Table 6 presents estimates for the probit regressions of job search channels. Only coefficients for the Covid status variable are reported, the base is the pre-Covid period. We also include the 'No search' variable, as defined in 2.A.3.

	Covid period	Post-covid	Controls	Time-trend
National employment agency	0.0090***	-0.0817***	√	√
Private employment agency	0.0462***	0.0863***	√	√
Personal network	0.0122***	-0.0157***	√	√
Job offers study	0.0665***	0.0562***	√	√
Competitive exam for the public sector	0.0331***	0.1030***	√	√
Test or job interview	0.0393***	0.1369***	√	√
Contact with an employer	-0.2793***	-0.4605***	√	√
Steps to become self-employed	0.2604***	0.4003***	√	√
Post of a job search/CV (inc, newspaper, online)	0.0448***	0.0122***	√	√
No search	-0.0332***	-0.0386***	√	√

Legend: * p<0.05; ** p<0.01; *** p<0.001. Only coefficients for the Covid status variable are reported, the pre-covid period is the base.

Table 6 – Probit estimates for the regression of job search channel use on covid status coefficients and control variables

To begin with, we observe a highly variable effect of Covid depending on the search channel, and in different directions. The two channels the most impacted are directed contact with an employer (the most negative coefficient) and the choice to become self-employed (the most positive coefficient). In both cases, the absolute value of the coefficient increases between the Covid period and the post-Covid period, showing a continuation and even an acceleration of the trend. Similarly, this continuation, although of a smaller magnitude, can also be seen in using a private employment agency, taking a competitive exam for the public sector, or doing a test or job interview. In these three cases, the search channel has been used more (positive coefficient) since the Covid period.

Conversely, some changes in the Covid period do not last and are of very low magnitude. Either the value of the coefficient decreases post-Covid (job offers study, post of a job search/CV), or there is even a change in the sign of the coefficient (personal network). We might have expected an increase in approaches that do not require direct interaction (post of a job search/CV, job offers

study), as opposed to direct contact with an employer. But this can be due to the way the variable is constructed since there is no distinction for entirely online procedures.

Finally, it appears that, since Covid, there has been no significant increase in the number of people not looking for work, beyond the time trend identified in Figure 2.

We now return to the determinants in more detail. We focus on the search channels for which, according to Table 6, the consequences of Covid are the strongest and consistent over time: test or job interview, contact with an employer, and steps to become self-employed. The *Competitive exam for the public secto*r variable is left aside because of its very low representation in our population (less than 2% of job-seekers say they used it in the last 4 weeks, see Table 2). Results are in Table 7. As these are probit regressions, we only comment on signs and magnitude of coefficients but we do not draw conclusions on exact marginal effects.

Interestingly, women do not take fewer tests or job interviews than men. For these channels do not observe any significant cross-effect of Covid, except with unemployment length. Unemployment length has a negative effect on taking a test or job interview in the pre-Covid period. This negative effect became more pronounced during Covid-19, with a greater impact the longer the duration of unemployment, which outweighs the overall positive effect. The situation seems to have improved slightly after the pandemic, but the negative impact remains significant. However, the use of tests or job interviews did not fall during Covid (see Table 6). This suggests the introduction of new practices, mainly remote. The negative effect observed could therefore reflect a discouragement on the part of the long-term unemployed in the face of these changes, or a period of adaptation needed to master digital tools and take account of changes in recruitment practices.

Variable	Test or job interview	Contact with an employer	Steps to become self-employed
Covid status			
Pre-Covid	(base)	(base)	(base)
Covid period	0.1738**	-0.3711***	0.4735***
Post-Covid	0.2711***	-0.6997***	0.4673***
Sexe			
Women	-0.018	0.0390***	-0.2886***
Women#Covid period	0.018	0.027	0.1305***
Women#Post-Covid	0.035	-0.021	0.1458**
Age range			
15-24 y.o	(base)	(base)	(base)
25-49 y.o.	-0.0887***	-0.2244***	0.3491***
50-64 y.o.	-0.2379***	-0.3195***	0.2931***
65 y.o. or more	-0.5808***	-0.5468***	0.095
25-49 y.o.#Covid period	0.015	-0.0619*	0.1273*
25-49 y.o.#Post-Covid	0.028	0.1699***	0.100
50-64 y.o.#Covid period	0.040	-0.035	0.003
50-64 y.o.#Post-Covid	0.001	0.2375***	0.016
65 y.o. or more #Covid period	-0.105	0.140	0.096
65 y.o. or more #Post-Covid	-0.4472*	0.166	-0.064
Education			
Primary education	(base)	(base)	(base)
Middle School	0.1886***	0.1193***	0.2175***
Vocational high school	0.2159***	0.1787***	0.2289***
High School	0.3705***	0.2531***	0.4664***
Two-year higher educa..	0.5620***	0.3304***	0.4970***
Bachelor degree or ab..	0.6225***	0.2570***	0.6179***
Middle School #Covid period	-0.043	0.021	-0.2843**
Middle School #Post-Covid	-0.125	0.040	-0.070
Vocational high…#Covid period	-0.006	-0.060	-0.2029**
Vocational high…#Post-Covid	-0.072	-0.005	-0.047
High School #Covid period	-0.010	0.002	-0.2886***
High School#Post-Covid	-0.060	-0.042	-0.1520*
Two-year higher... #Covid period	-0.1425**	-0.1413**	-0.1441*
Two-year higher... #Post-Covid	-0.1450*	-0.031	-0.022
Bachelor degree...#Covid period	-0.006	-0.021	-0.1631*
Bachelor degree...#Post-Covid	-0.1301*	0.039	-0.114
Family setup			
Single person	(base)	(base)	(base)
Single parent family	-0.0452*	-0.014	0.010
Couple without children	-0.011	-0.004	0.043
Couple with children	-0.0482**	0.013	0.026
Other	-0.0670**	-0.0735***	0.013
Single parent…#Covid period	-0.021	-0.035	-0.056
Single parent…#Post-Covid	0.042	-0.033	-0.093
Couple without…#Covid period	-0.039	-0.019	0.042
Couple without…#Post-Covid	0.042	-0.1150*	0.031
Couple with ch…#Covid period	-0.032	-0.0744*	0.014
Post-Covid	0.010	-0.077	0.052
Other#Covid period	-0.087	-0.034	0.049
Other#Post-Covid	0.034	-0.100	-0.168

Unemployment length			
Less than 3 months	(base)	(base)	(base)
3-6 months	0.016	0.1131***	0.1041**
6-12 months	-0.0393*	0.1330***	0.1826***
12-24 months	-0.1356***	0.0808***	0.1271***
2-4 years	-0.2095***	0.034	0.1048**
More than 4 years	-0.4095***	-0.1217***	0.000
3-6 months#Covid period	-0.066	0.005	-0.082
3-6 months#Post-Covid	-0.080	-0.0956*	0.038
6-12 months#Covid period	-0.1120**	-0.037	-0.095
6-12 months#Post-Covid	-0.1273**	-0.1409**	0.125
12-24 months#Covid period	-0.0827*	0.038	-0.077
12-24 months#Post-Covid	-0.1395**	-0.1373**	0.096
2-4 years#Covid period	-0.2516***	-0.061	-0.097
2-4 years#Post-Covid	-0.1830**	-0.1760**	-0.064
More than 4 years#Covid period	-0.2019***	0.075	-0.067
More than 4 years#Post-Covid	-0.2005*	-0.080	0.031
Previous job experience			
No	-0.0953***	-0.0606***	-0.3723***
No#Covid period	-0.061	-0.024	0.1788*
No#Post-Covid	0.038	0.012	0.132
Constant	-0.9168***	0.2996***	-2.4702***
R^2	0.047	0.046	0.088
N	166,416	166,416	166,416

Legend: * p<0.05; ** p<0.01; *** p<0.001

Table 7 – Probit estimates for the regression of 3 job search channel use on covid status coefficients and control variables, allowing for cross-effects

The evolution of recruitment practices with the development of online tools is an ongoing field of research both in terms of the quality of matching from the employer's point of view and accessibility for job-seekers. Familiarity with digital tools and access to a reliable internet connection is already an obstacle for some candidates [see for example Vashistha et al., 2021, who investigate the online recruitment and the use of web-based technology in the hiring process in IT Companies of Noida Region in Turkey].

The effect of unemployment length also changes for direct contact with an employer. A duration of unemployment between 3 and 24 months had a positive effect on the use of this channel pre-Covid, with positive effects of the same order of magnitude (coefficients between 0.0808 and 0.1330). On the other hand, in the post-Covid period, there is an increasing negative effect of unemployment length. Conversely, the negative effect of the job-seeker's age on contact

with an employer is attenuated since the end of Covid (see coefficients for 25-49 y.o.#Post-Covid and 50-64 y.o.#Post-Covid, significant at the 0.1% level).

Finally, gender effect on self-employment choice has been reduced by half since the Covid period. A similar observation is made by Cai and Baker [2023] in the United States. The authors show a substantial increase in self-employment, a trend they expect to continue in the long term. Above all, this US increase is disproportionate among women, particularly those with young children at home. Here, we also observe a negative shock to education coefficients during the Covid period, but this is no longer significant in the post-pandemic period.

To conclude, there are no overall big cross-effects. Determinants seem to be more or less the same before and after Covid, with some transitional changes during the Covid period because the same actions were not available. There are two exceptions to this result. Firstly, the negative effect of the duration of unemployment on direct contact with an employer and on tests or interviews. Secondly, a reduction in the gap between men and women in the decision to become self-employed. These two trends appear to be persisting.

4. Conclusion

In this essay, we tried to assess some changes in job search behaviours following the Covid pandemic, by studying unemployed people's choice of job search channels. Here are our main findings. To begin with, there is a general trend towards a reduction in active job search, which has been associated since the Covid period with a reduction in the number of channels used (lower search intensity). There is little effect of the pandemic on the determinants of job search intensity, apart from a reduction in gender bias and a lower magnitude of the post-Covid coefficients.

In terms of job search content, Covid has the most positive effect on the use of test or job interview and steps to become self-employed. Conversely, we logically observe a sharp decline in

contact with an employer, but this seems to be confirmed even after the pandemic. A more detailed study of the determinants does not reveal any significant cross-effects. However, it raises a concern about the adaptation of people who have been unemployed for a long time to changes in practices. We also note the higher proportion of women who choose to become self-employed.

This essay is constrained by its length and the data available, but we feel that there are several interesting directions for future research.

First and foremost, our data hides some of the possible changes. Three angles seem to be particularly lacking in our work. Firstly, the job search channels we have here do not reveal Internet-related practices, even though the digital divide (connection, skills) can become a real handicap when looking for a job. Similarly, there is no information on job searches via online social networks (particularly Linkedin), which seems to be developing quickly in some employment sectors. Secondly, our data lacks temporal perspective, as the Covid pandemic is still very recent. Above all, we do not have enough years to expand on the post-Covid period and strengthen our conclusions on the use of job search channels over the long term. Thirdly, spatial constraints seem to be essential in the choice of these search channels [Bouabdallah et al., 2002], and are likely to change depending on the sector following the development of remote technology.

Moreover, we mentioned that job-seekers are likely to have different job search behaviour whether they are currently employed or not. A reasonable hypothesis is that job-seekers who are already employed are better informed about changes in the job market and recruitment following the pandemic (dynamic sectors, changes in practice, numeric tools). Studying changes in their job search practices would enable us to see not only whether the unemployed have changed their job search behaviour compared with the pre-Covid period, but above all whether they have moved away from what the non-unemployed do in their choice of search channels. Such a comparison, beyond the scope of this essay, would definitely deepen the analysis.

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